

Molecular-Subtype-Specific Biomarkers Improve Prediction of Prognosis in Colorectal Cancer.

PMID: 28494874 | Indexed in PubMed

Colorectal cancer (CRC) is characterized by major inter-tumor diversity that complicates the prediction of disease and treatment outcomes. Recent efforts help resolve this by sub-classification of CRC into natural molecular subtypes; however, this strategy is not yet able to provide clinicians with improved tools for decision making. We here present an extended framework for CRC stratification that specifically aims to improve patient prognostication. Using transcriptional profiles from 1,100 CRCs, including >300 previously unpublished samples, we identify cancer cell and tumor archetypes and suggest the tumor microenvironment as a major prognostic determinant that can be influenced by the microbiome. Notably, our subtyping strategy allowed identification of archetype-specific prognostic biomarkers that provided information beyond and independent of UICC-TNM staging, MSI status, and consensus molecular subtyping. The results illustrate that our extended subtyping framework, combining subtyping and subtype-specific biomarkers, could contribute to improved patient prognostication and may form a strong basis for future studies.

Targeting HDAC6 improves anti-CD47 immunotherapy.

PMID: 38414061 | Indexed in PubMed

Cancer cells can overexpress CD47, an innate immune checkpoint that prevents phagocytosis upon interaction with signal regulatory protein alpha (SIRP α) expressed in macrophages and other myeloid cells. Several clinical trials have reported that CD47 blockade reduces tumor growth in hematological malignancies. However, CD47 blockade has shown modest results in solid tumors, including melanoma. Our group has demonstrated that histone deacetylase 6 inhibitors (HDAC6is) have immunomodulatory properties, such as controlling macrophage phenotype and inflammatory properties. However, the molecular and cellular mechanisms controlling these processes are not fully understood. In this study, we evaluated the role of HDAC6 in regulating the CD47/SIRP α axis and phagocytosis in macrophages. We tested the role of HDAC6is, especially Nexturastat A, in regulating macrophage phenotype and phagocytic function using bone marrow-derived macrophages and macrophage cell lines. The modulation of the CD47/SIRP α axis and phagocytosis by HDAC6is was investigated using murine and human melanoma cell lines and macrophages. Phagocytosis was evaluated via coculture assays of macrophages and melanoma cells by flow cytometry and immunofluorescence. Lastly, to evaluate the antitumor activity of Nexturastat A in combination with anti-CD47 or anti-SIRP α antibodies, we performed in vivo studies using the SM1 and/or B16F10 melanoma mouse models. We observed that HDAC6is enhanced the phenotype of antitumoral M1 macrophages while decreasing the protumoral M2 phenotype. In addition, HDAC6 inhibition diminished the expression of SIRP α , increased the expression of other pro-phagocytic signals in macrophages, and downregulated CD47 expression in mouse and human melanoma cells. This regulatory role on the CD47/SIRP α axis translated into enhanced antitumoral phagocytic capacity of macrophages treated with Nexturastat A and anti-CD47. We also observed that the systemic administration of HDAC6i enhanced the in vivo antitumor activity of anti-CD47 blockade in melanoma by modulating macrophage and natural killer cells in the tumor microenvironment. However, Nexturastat A did not enhance the antitumor activity of anti-SIRP α despite its modulation of macrophage populations in the SM1 tumor microenvironment. Our results demonstrate the critical regulatory role of HDAC6 in phagocytosis and innate immunity for the first time, further underscoring the use of these inhibitors to potentiate CD47 immune checkpoint blockade therapeutic strategies.

CD47 regulation of epithelial cell spreading and migration, and its signal transduction.

PMID: 16776778 | Indexed in PubMed

CD47 is an integrin-associated penta-transmembrane protein that possesses an immunoglobulin-like domain in its extracellular region. We have now investigated the role of CD47 in the regulation of epithelial cell spreading and migration. CD47 is colocalized with E-cadherin at cell-cell adhesion sites of epithelial cells. A Ca^{2+} switch experiment showed that CD47 was endocytosed and then relocalized to cell-cell adhesion sites in a similar manner to E-cadherin. Such polarized localization of CD47 required the multiple spanning region of this protein. Forced expression of CD47 induced cell spreading with marked lamellipodium formation and resulted in both partial disruption of cell-cell adhesion and enhancement of the hepatocyte growth factor-stimulated scattering of Madin-Darby canine kidney cells. The CD47-induced cell spreading was blocked by inhibition of Src and mitogen-activated protein kinase kinase. Thus, these results suggest that CD47 participates in the regulation of cell-cell adhesion and cell migration through reorganization of the actin cytoskeleton in epithelial cells. This function of CD47 is mediated by the activation of Src and mitogen-activated protein kinase kinase.

[Research on physiological function of CD47].

PMID: 12962580 | Indexed in PubMed

CD47, also known as integrin-associated protein (IAP), is an ubiquitously expressed 50 kD cell membrane glycoprotein. This article reviews that as an integrin-associated molecule, CD47 regulates a series of integrin-dependent cell functions; CD47 regulates cell adhesiveness, migration and activation and platelet aggregation through binding to thrombospondins; CD7 is the extracellular ligand for human signal-regulatory proteins, thus it causes negative regulatory effect; CD47 induces costimulatory signals on activation of T cell, T cell apoptosis and T cell anergy and enhances the efficiency of TCR signaling; CD47 is a member of Rh antigen complex. CD47 is involved in stroma-supported erythropoiesis. This review also discussed that CD47 as a marker on red blood cells, associated with haemolysis, which provides a new way to recognize, diagnose, and treat diseases.

CD47 update: a multifaceted actor in the tumour microenvironment of potential therapeutic interest.

PMID: 22774848 | Indexed in PubMed

CD47 is a ubiquitous 50 kDa five-spanning membrane receptor that belongs to the immunoglobulin superfamily. This receptor, also known as integrin-associated protein, mediates cell-to-cell communication by ligation to transmembrane signal-regulatory proteins SIRP α and SIRP γ and interacts with integrins. CD47 is also implicated in cell-extracellular matrix interactions via ligation with thrombospondins. Furthermore, CD47 is involved in many and diverse cellular processes, including apoptosis, proliferation, adhesion and migration. It also plays a key role in many immune and cardiovascular responses. Thus, this multifaceted receptor might be a central actor in the tumour microenvironment. Solid tumours are composed of not only cancer cells that actively proliferate but also

other cell types including immune cells and fibroblasts that make up the tumour microenvironment. Tumour cell proliferation is strongly sustained by continuous sprouting of new vessels, which also represents a gate for metastasis. Moreover, infiltration of inflammatory cells is observed in most neoplasms. Much evidence has accumulated indicating that infiltrating leukocytes promote cancer progression. Given its ubiquitous expression on all the different cell types that compose the tumour microenvironment, targeting CD47 could represent an original therapeutic strategy in the field of oncology. We present a current overview of the biological effects associated with CD47 on cancer cells and stromal cells.

CD47 signaling pathways controlling cellular differentiation and responses to stress.

PMID: 25708195 | Indexed in PubMed

CD47 is a widely expressed integral membrane protein that serves as the counter-receptor for the inhibitory phagocyte receptor signal-regulatory protein- α (SIRP α) and as a signaling receptor for the secreted matricellular protein thrombospondin-1. Recent studies employing mice and somatic cells lacking CD47 have revealed important pathophysiological functions of CD47 in cardiovascular homeostasis, immune regulation, resistance of cells and tissues to stress and chronic diseases of aging including cancer. With the emergence of experimental therapeutics targeting CD47, a more thorough understanding of CD47 signal transduction is essential. CD47 lacks a substantial cytoplasmic signaling domain, but several cytoplasmic binding partners have been identified, and lateral interactions of CD47 with other membrane receptors play important roles in mediating signaling resulting from the binding of thrombospondin-1. This review addresses recent advances in identifying the lateral binding partners, signal transduction pathways and downstream transcription networks regulated through CD47 in specific cell lineages. Major pathways regulated by CD47 signaling include calcium homeostasis, cyclic nucleotide signaling, nitric oxide and hydrogen sulfide biosynthesis and signaling and stem cell transcription factors. These pathways and other undefined proximal mediators of CD47 signaling regulate cell death and protective autophagy responses, mitochondrial biogenesis, cell adhesion and motility and stem cell self-renewal. Although thrombospondin-1 is the best characterized agonist of CD47, the potential roles of other members of the thrombospondin family, SIRP α and SIRP γ binding and homotypic CD47 interactions as agonists or antagonists of signaling through CD47 should also be considered.

Novel CD47-dependent intercellular adhesion modulates cell migration.

PMID: 15880429 | Indexed in PubMed

CD47 is a ubiquitously expressed plasma membrane protein, also known as Integrin Associated Protein, that modulates cell adhesion both through alteration of the avidity of integrin binding and through interaction with its own ligands, the extracellular matrix protein thrombospondin (TSP) and the plasma membrane response regulator SIRP α 1. We now show that CD47 expression on fibroblasts can induce intercellular adhesion resulting in cell aggregation in the absence of active integrins, SIRP α 1 binding, and detectable TSP. CD47-expressing cells preferentially bind to other CD47-expressing cells, and intercellular adhesion requires stimulation by serum or a CD47-binding peptide from TSP. Cell-cell adhesion is inhibited by pertussis toxin and C. difficile toxin B, and both adherent and aggregating CD47-expressing fibroblasts have more rac in the GTP bound state than CD47-deficient cells. Spontaneous migration of Jurkat lymphocytes through a fibroblast monolayer is decreased by fibroblast

expression of CD47, consistent with an increased barrier function of the CD47 expressing cells. The lymphocyte chemoattractant SDF-1 α stimulates migration of Jurkat cells through this monolayer only if both the lymphocytes and fibroblasts express CD47, and the inhibition of migration by a CD47-interacting peptide from TSP similarly requires CD47 expression on both cell types. Thus, signaling dependent on both heterotrimeric and rho family GTPases can induce CD47 to participate in cell-cell interactions independent of known ligands that enhance intercellular adhesion and modulate cell migration.

Role of CD47 in erythroid cells and in autoimmunity.

PMID: 15359629 | Indexed in PubMed

The cell surface glycoprotein CD47 (Integrin-associated protein/IAP) was originally identified as a regulator of integrin-dependent responses to extracellular matrix proteins. However, CD47 is ubiquitously expressed, also by cells that do not express integrins. Thus, during the last few years, it has been shown that CD47 has several important functions besides assisting integrin activation. This review will focus on the role of CD47 in erythrocytes. In these cells, CD47 was found to be an important link in the interaction between the band 3 complex and the Rh complex in the maintenance of erythrocyte membrane integrity. CD47 can also function as a marker of self on erythrocytes, and likely also on other cells, by binding to the inhibitory receptor SIRP α . In this way, SIRP α -expressing cells, like macrophages and dendritic cells, are less likely to phagocytose an autoimmune sensitized cell with CD47 on its surface than a CD47-deficient cell where this inhibitory mechanism will not be engaged. The interaction between CD47 and SIRP α seems to be important to limit destruction of host cells in autoimmune diseases like autoimmune hemolytic anemia (AIHA), where macrophages destroy antibody or complement opsonized cells.

CD47: A Cell Surface Glycoprotein Which Regulates Multiple Functions of Hematopoietic Cells in Health and Disease.

PMID: 23401787 | Indexed in PubMed

Interactions between cells and their surroundings are important for proper function and homeostasis in a multicellular organism. These interactions can either be established between the cells and molecules in their extracellular milieu, but also involve interactions between cells. In all these situations, proteins in the plasma membranes are critically involved to relay information obtained from the exterior of the cell. The cell surface glycoprotein CD47 (integrin-associated protein (IAP)) was first identified as an important regulator of integrin function, but later also was shown to function in ways that do not necessarily involve integrins. Ligation of CD47 can induce intracellular signaling resulting in cell activation or cell death depending on the exact context. By binding to another cell surface glycoprotein, signal regulatory protein alpha (SIRP α), CD47 can regulate the function of cells in the monocyte/macrophage lineage. In this spotlight paper, several functions of CD47 will be reviewed, although some functions may be more briefly mentioned. Focus will be on the ways CD47 regulates hematopoietic cells and functions such as CD47 signaling, induction of apoptosis, and regulation of phagocytosis or cell-cell fusion.

[Role of CD47 in hematologic malignancies].

PMID: 24370062 | Indexed in PubMed

CD47 is a ubiquitously expressed transmembrane glycoprotein on surface of many cells. Through its interaction with integrin, signal regulatory protein alpha (SIRP α) and thrombin sensitive protein-1 (TSP-1), it plays important roles in various immunological processes including inflammatory response, immune response and tumor immunity. Recently, it has been found that CD47 interacts with SIRP α expressed on phagocytic cells, which transfers a negative signal when being activated. By the mechanisms described above, CD47-SIRP α signal complex is involved in the pathogenesis of hematological diseases and might provide some informations for the therapy of patients. This review focuses on the structure and immunoregulatory functions of CD47, the mechanism of CD47 in tumor therapy, the CD47 and hematologic malignancies including acute leukemia, B-cell lymphoma and multiple myeloma, as well as CD47 and hematopoietic stem cell transplantation.
